AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 4/Dec. /2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90 Minutes

Instructions:

- Total Marks: **30**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple Choice Questions: [5 marks]

Question 1: Choose the most correct answer.

1. Which among the following is accommodated by the **main memory**?
   a) Currently executing programs
   b) Data currently in use
   c) Operating system
   d) All the above

2. Which of the following is **not** a system call?
   a) open()
   b) read()
   c) printf()
   d) None of the above

3. Which of the following is **not true** about cloud computing?
   a) It allows on-demand access to computing resources over the internet
   b) It eliminates the need for physical data centers and servers
   c) It guarantees 100% uptime with no risk of failure
   d) None of the above

4 An example of **IaaS** is:
   a) Google Docs
   b) Dropbox
   c) MS Excel
   d) Amazon EC2 server

5. Which of the following is **not related** to the kernel?
   a) User interface
   b) Process management
   c) Memory management
   d) All the above

Part 2: Short Essay Questions: [15 marks]

Question 2: [2 marks]
What is Middleware? Define

Question 3: [2 marks]
Multiprogramming offers several advantages, including increased CPU utilization and improved memory utilization. Briefly explain how multiprogramming achieves each of these benefits.

Question 4: [3 marks]
What do the following acronyms stand for?
   a- GUI
   b- API
   c- SAN

Question 5: [4 marks]
Answer the following questions:
a) Define concurrency and parallelism.
b) Which of the two—concurrency or parallelism—can occur on a single-core system and on a multi-core system?

Question 6: [4 marks]
   a) What are the different process states? (2.5 marks)
   b) Draw process state diagram. (1.5 marks)

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Part 3: Problem Solving: [10 marks]

Question 7: [4 marks]

a) What is the meaning of the following crontab time format? (2 marks)

15 7-9 * * *

b) What does ***chmod*** do? Provide an example. (2 marks)

Question 8: [6 marks]

Write a **Bash script** that:

1. Prompts the user to **enter the number of temperature readings** to process.
2. Reads each temperature (in Celsius) one by one.
3. Counts how many temperatures fall into each category:
   - **Hot:** temperature > 30°C
   - **Moderate:** temperature between 15°C and 30°C (inclusive)
   - **Cold:** temperature < 15°C
4. After all temperatures have entered, display the count of hot, moderate, and cold days.

Remark: *Assume all temperature inputs are valid numbers.*

Expected Outputs (Example Run):

Enter the number of temperature readings: 3
Enter temperature 1: 32
Enter temperature 2: 15
Enter temperature 2: 30

Number of Hot days: 1
Number of Moderate days: 2
Number of Cold days: 0

End of Question

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